

CLOUD PLATTER

A Project Report

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in partial fulfillment for the award of the degree

of

**MASTER OF SCIENCE – INFORMATION
TECHNOLOGY**



Department of Computer Science

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Acknowledgement

We would like to extend our heartfelt gratitude to everyone who has contributed to the successful completion of our project, “Cloud Platter.” This project is the result of the collective effort, dedication, and guidance of many individuals, and we deeply appreciate their invaluable support throughout this journey.

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Our heartfelt thanks also go to Indus University for providing us with the necessary resources, technical facilities, and an environment conducive to learning and innovation.

Finally, we wish to express our sincere appreciation to our teammates, friends, and family members for their cooperation, patience, and encouragement. Each contribution—whether direct or indirect—has been instrumental in bringing Cloud Platter to life.



CERTIFICATE

This is to certify that the project titled **“CLOUD PLATTER”** is the bonafide work carried out by **Shah Tisha KetanKumar (IU2453000112) , Doshi Drashti Atmesh ((IU2453000124) , Aghera Nensi Rasikbhai (IU2453000126)** a student of MCA SEM - III during the academic year 2025 - 26, in partial fulfillment of the requirements for the award of the degree of Master of Computer Application.

Internal Guide Signature :

Head of Dept. Signature :

Date : 15/10/2025

Date :15/10/2025

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INTRODUCTION

- We are offering a seamless ordering experience, comprehensive menu options, and robust back-end management capabilities, ultimately driving customer satisfaction, operational efficiency, and revenue growth.
- An objective for developing a cloud kitchen website is to create a user-friendly online platform that efficiently connects customers with our virtual kitchen services.
- The user experience of our food ordering website aimed at enhancing customer satisfaction and operational efficiency. In today's fast-paced world, online food ordering platforms play a crucial role in providing convenience and accessibility to customers while streamlining processes for restaurants.
- Further more, the website's performance in terms of speed, reliability, and scalability. It explores strategies for optimizing loading times, minimizing downtime, and accommodating fluctuations in user traffic to ensure seamless operation during peak hours.
- Moreover, the report examines the website's compatibility with various devices and browsers to ensure accessibility across different platforms. It also discusses the implementation of responsive design principles to create a consistent experience across desktops, tablets, and smartphones.

Existing system

- A cloud kitchen is a delivery-only restaurant model that operates without a dine-in facility. The existing system typically involves optimized workflows, manage food preparation, order processing, delivery.
- Minimal front-of-house staff.
- Updates inventory and order status.
- In-house delivery staff .

Problem areas of existing system

- Food may spill, get cold, or lose quality during delivery.
- Increased packaging costs to ensure safety and hygiene.
- Delivery Issues:-Delays, customer complaints
- **Limited personalization:** The system does not provide personalized menu suggestions or recommendations.
- **Manual processes:** Order management and tracking appear basic, with no real-time optimization or monitoring.

Need for new System

- Limited access to customer data and no personalized marketing.
- To stay competitive, the system needs continuous improvements in usability, scalability, and customer engagement.
- To handle future growth, the platform must expand beyond basic ordering into smarter, automated, and transparent systems.

Proposed System

- 1.User Registration/Login** – Users create an account or log in.
- 2.View Menu** – Browse food categories and items.
- 3.Add to Cart** – Select and add items to the cart.
- 4.Checkout** – Review order and proceed to payment.
- 5.Online Payment** – Pay securely via **Stripe**.
- 6.Order Confirmation** – System confirms and saves the order.
- 7.Admin Panel** – Admin manages menu, orders, and users.
- 8.Feedback & Contact** – Users give feedback or send queries.
- 9.Database Storage** – All data securely stored in **MongoDB**.

Module & user

cloud platform a user-friendly website for a cloud kitchen, integrating online ordering, menu management, for seamless customer experience.

1. Admin Module

The **Admin module** provides complete control over the restaurant system. The administrator is responsible for managing data, supervising operations, and ensuring smooth workflow.

Key Features:

- Manage customer details (view, update, remove)
- Add or delete menu products
- Approve or reject customer orders

2. Customer Module

The **Customer module** is designed to provide a user-friendly dining experience. It allows customers to interact with the platform easily and securely.

Key Features:

- Account registration and secure login
- Browse menu items with categories and details
- Place orders online with real-time updates
- Manage account details and preferences
- View past orders and order history
- Provide feedback and reviews for products and services

Tools & Technology

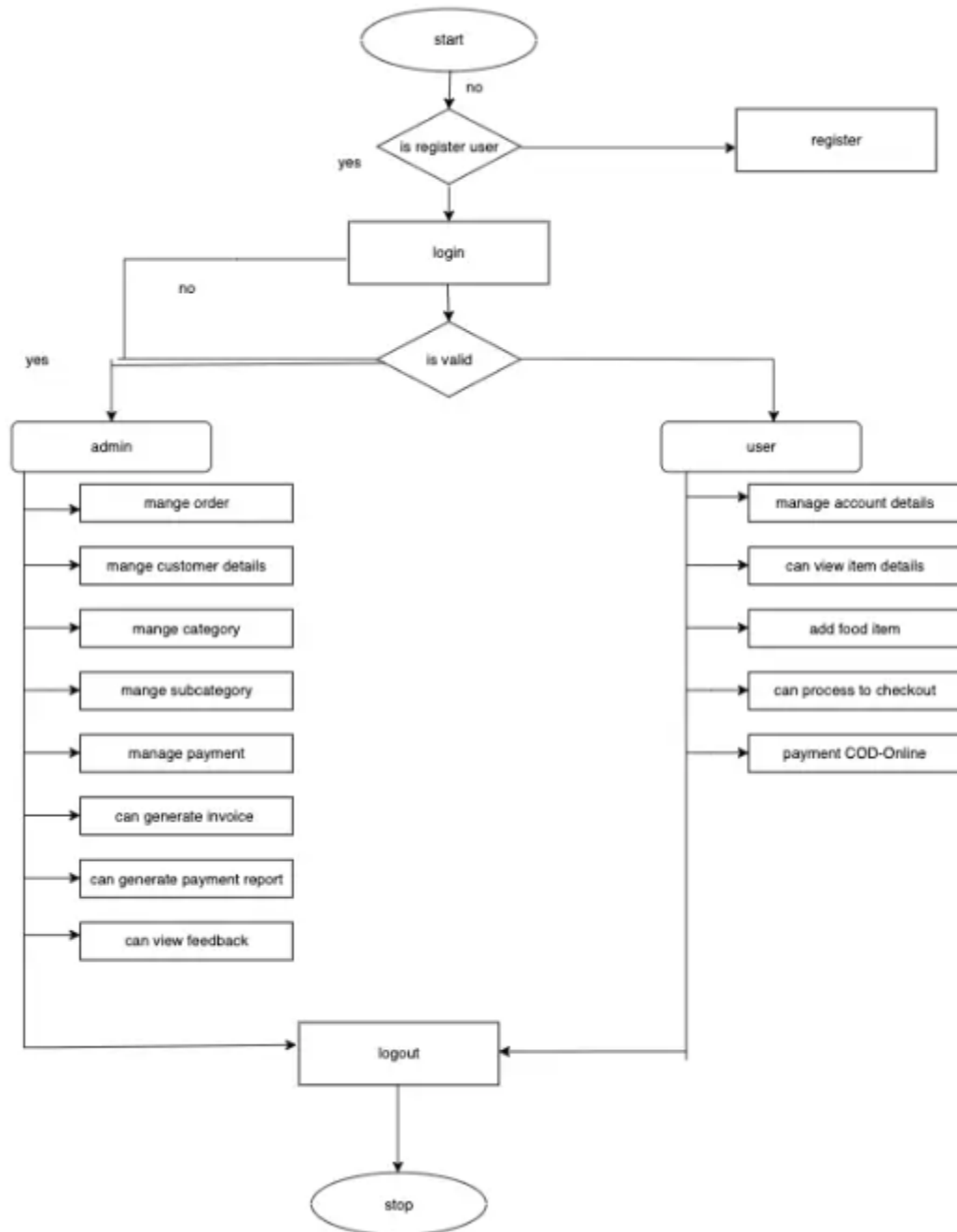
Objective: Developing a captivating and user-friendly restaurant website to enhance online visibility, streamline menu access, and provide a seamless dining experience for customers.

Platform : Visual Studio Code

Front End Tool: React.js 19.0.0

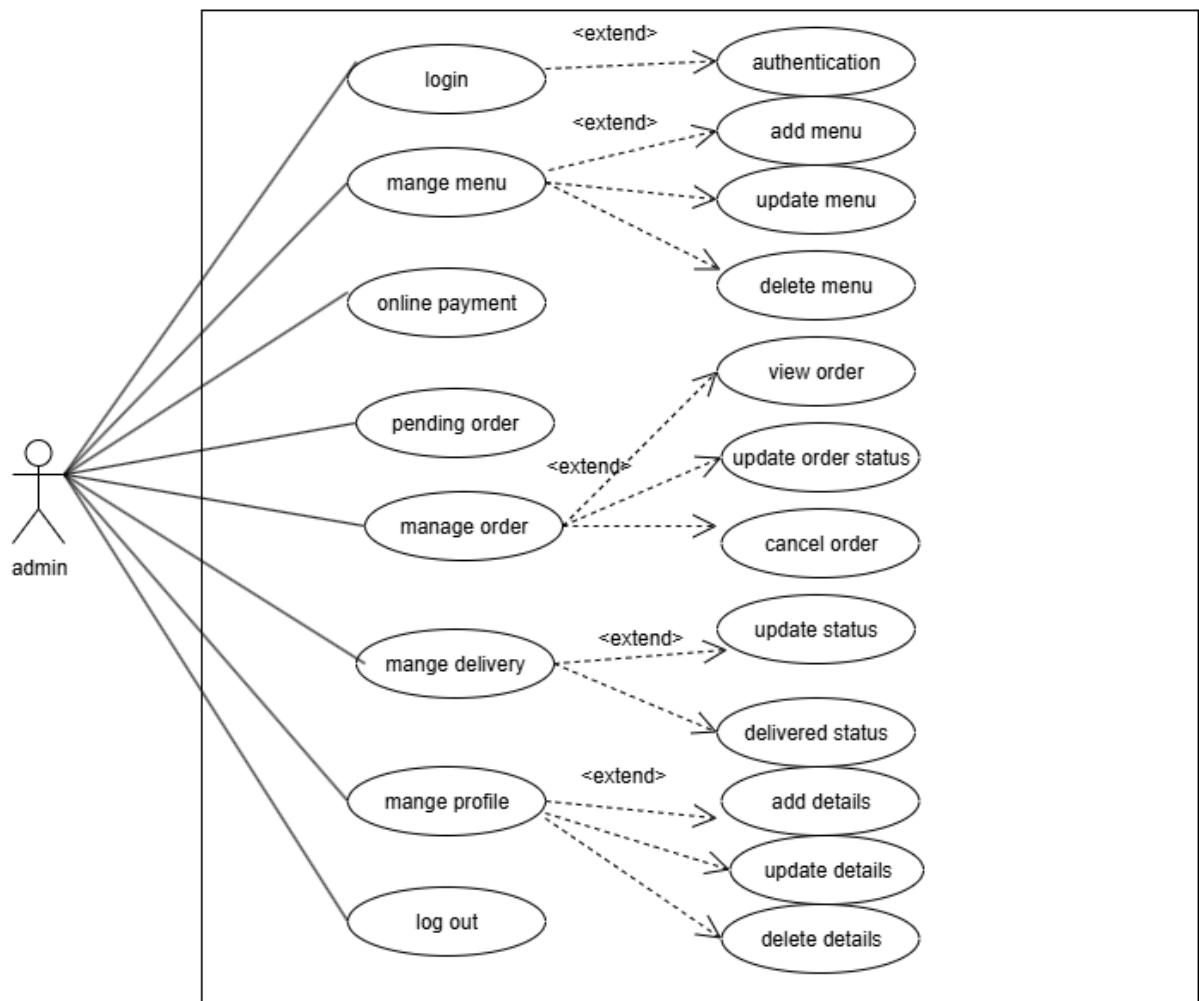
Back End Tool : Node.js 24.6.0 , Express.js 5.1.0, MongoDB 8.0, Stripe

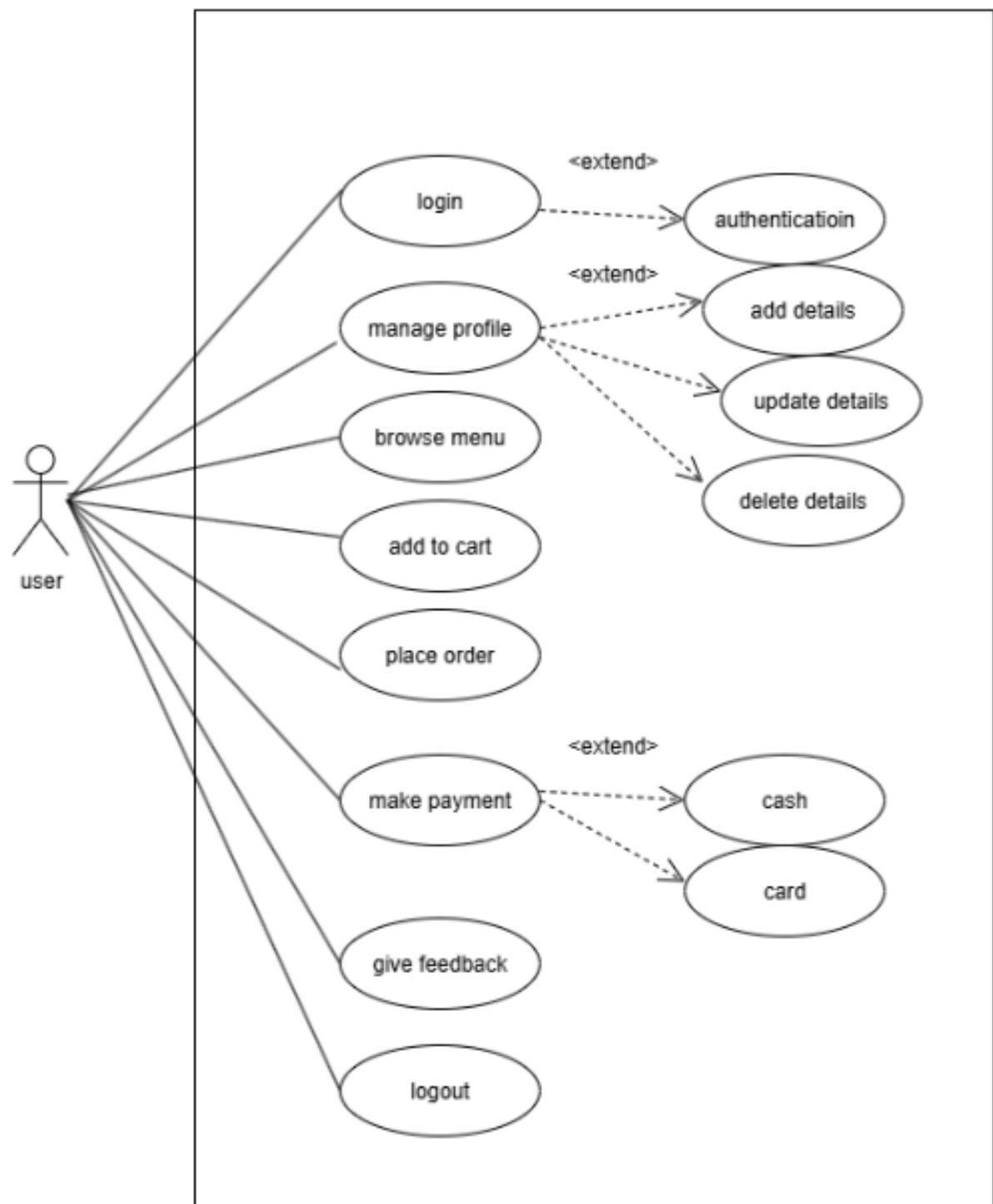
System Flow Diagram



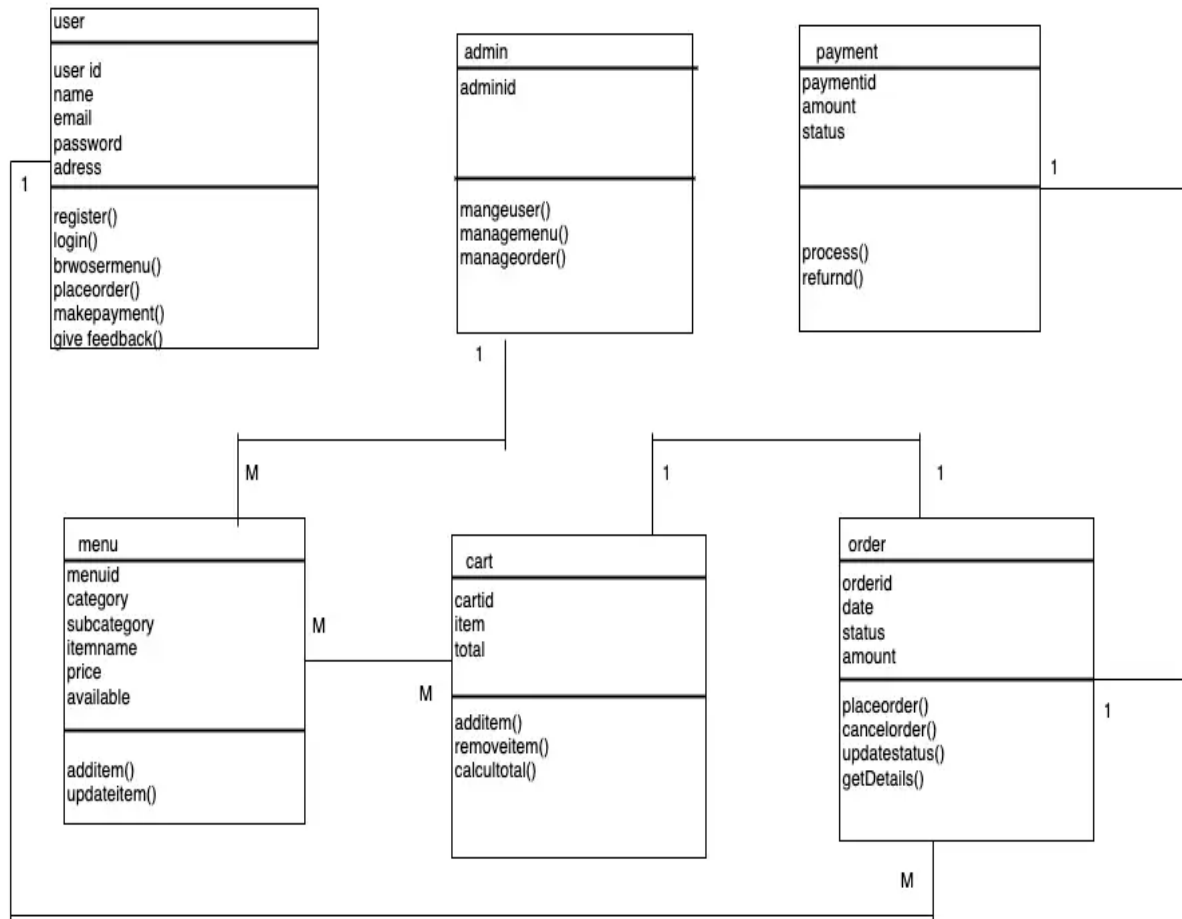
use case diagram

Admin:



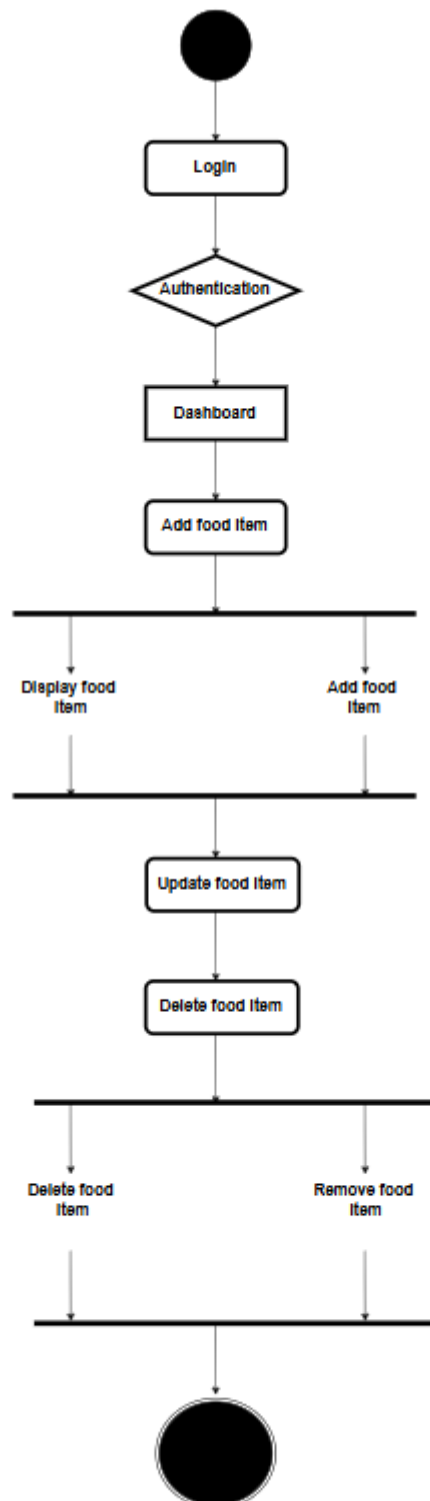
User:

Class Diagrams

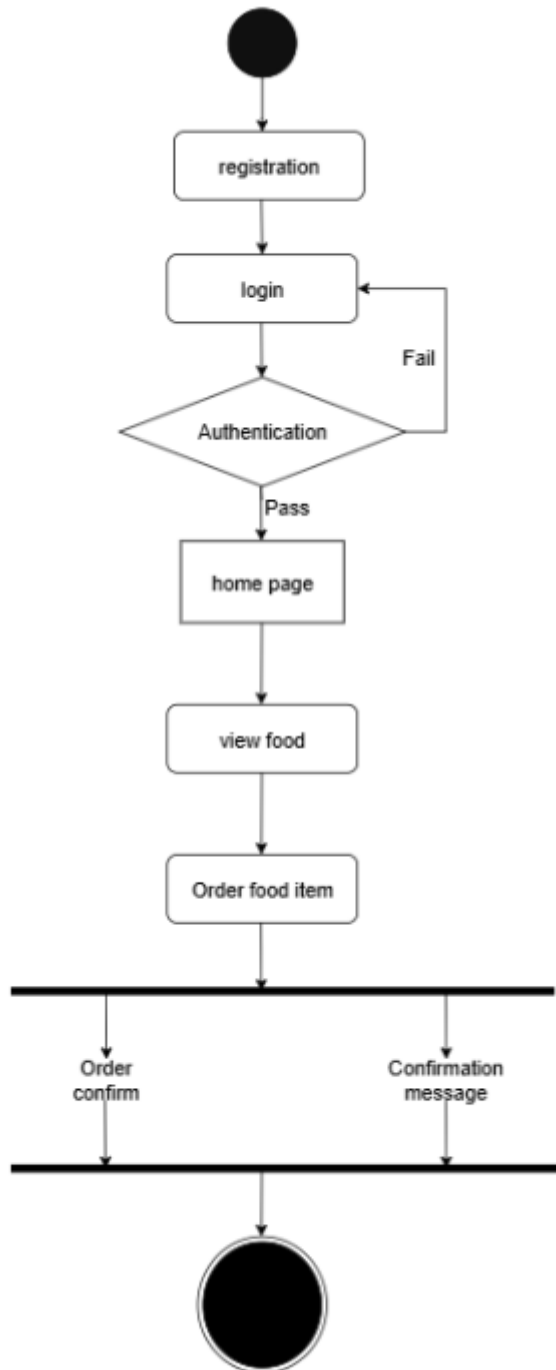


Activity Diagrams

Admin

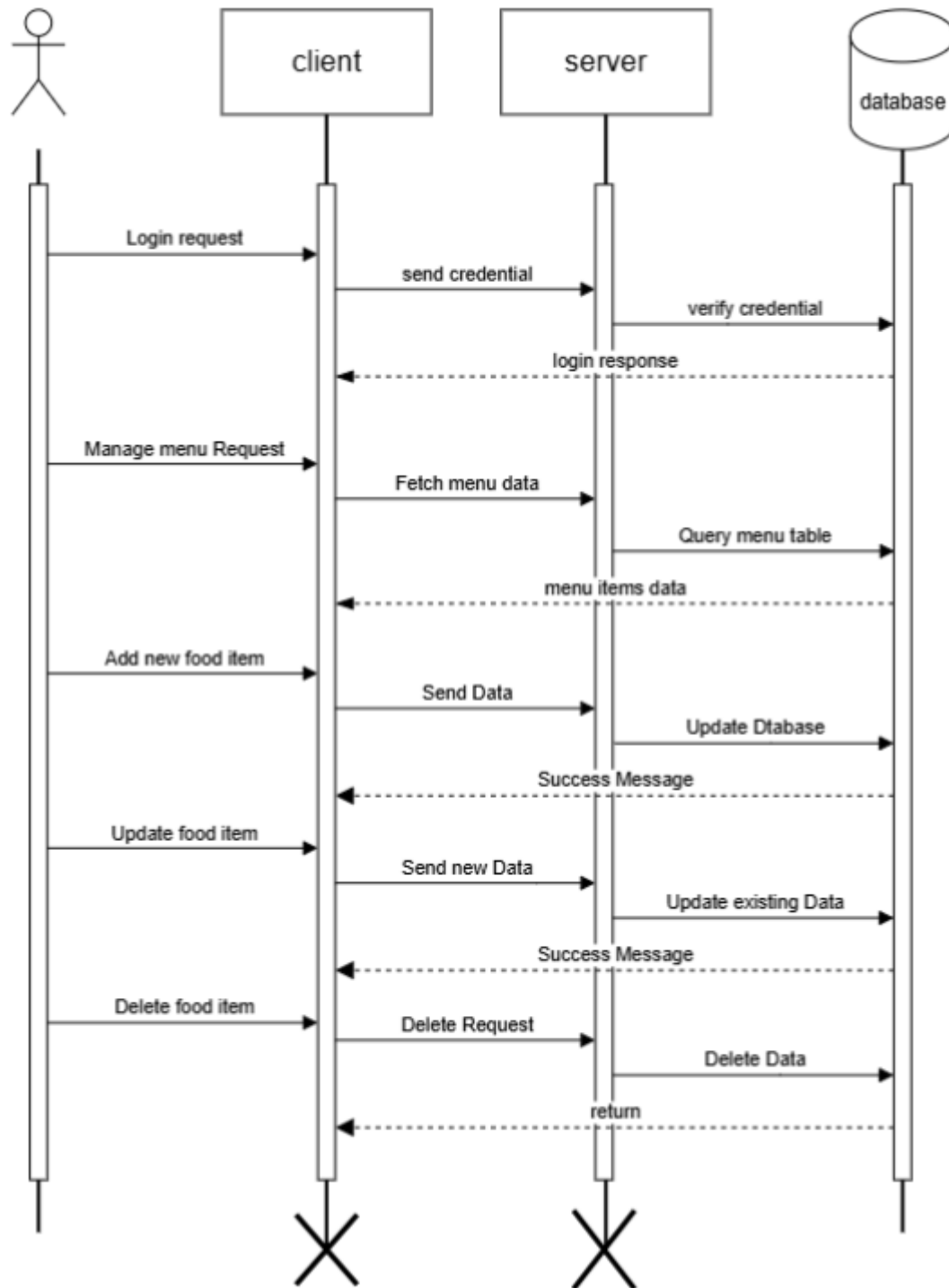


User:

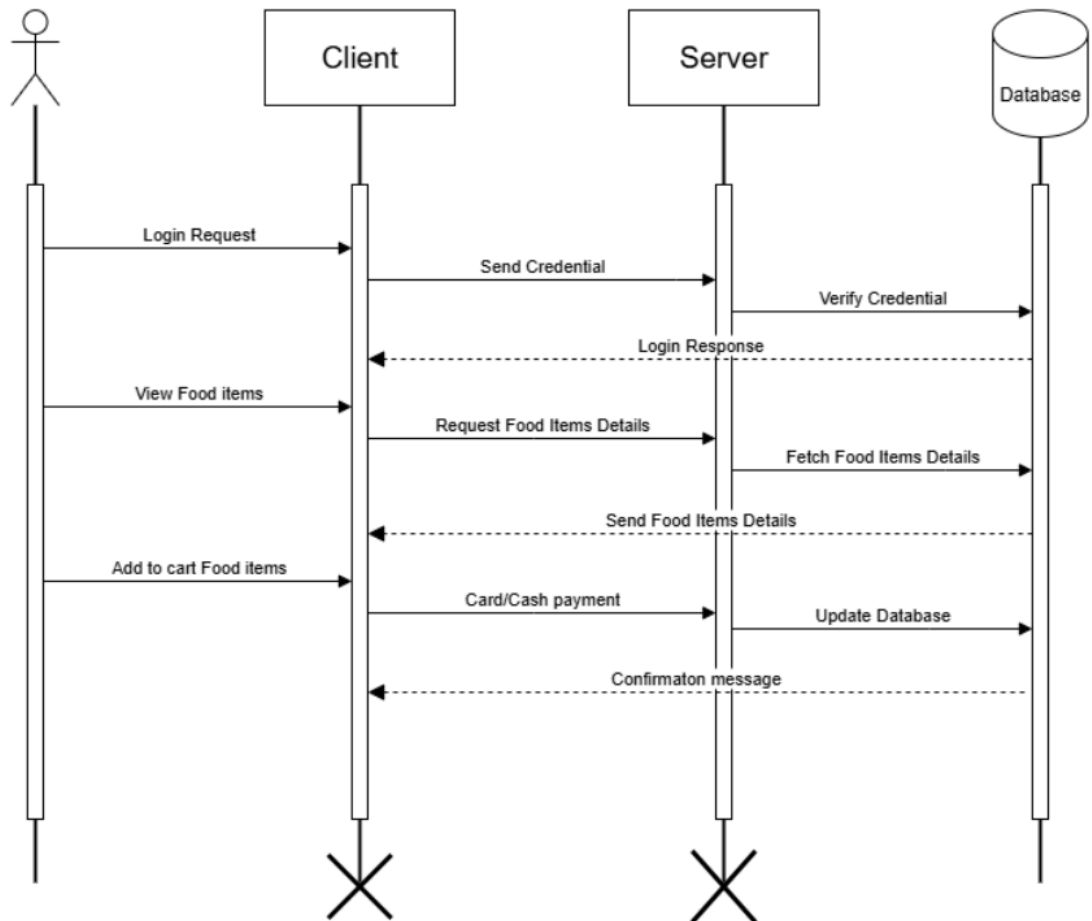


Sequence Diagrams

Admin:



user:



Data Dictionary

Table Name : Admin

Description : It is used to store all Admin detail

Fieldname	Data type	Size	Constraint	Description
Admin_id	Int	10	Primary Key	To Store Admin Id .
Admin_name	varchar	30	Not null	To Store Admin Name.
Email_id	varchar	30	Not null	To Store Email Id.
Password	varchar	15	Not null	To Store Password

Table Name : SIGN-UP

Description : It is used to store registration details

Fieldname	Data type	Size	Constraint	Description
User_id	Int	10	Primary Key	To Store User Id .
Username	varchar	30	Not null	To Store User Name.
Email_id	varchar	30	Not null	To Store Email Id.
Password	varchar	15	Not null	To Store Password.

Table Name : Item_Category

Description : It is used to store Item details

Fieldname	Data type	Size	Constraint	Description
Category_id	Int	10	Primary Key	To Store Category Id .
Category name	varchar	30	Not null	To Store Category Name.
Description	varchar	50	Not null	To Store Description.

Table Name : Item_subcategory

Description : It is used to store Item subcategory details

Fieldname	Data type	Size	Constraint	Description
Subcategory_id	Int	10	Primary Key	To Store Subcategory Id .
Category_Id	Int	30	Foriegn Key	To Store Category Id.
Subcategory - name	varchar	30	Not null	To Store Subcategory Name.
Subcategory - description	varchar	30	Not null	To Store Subcategory Description

Table Name : Cart_Page**Description : It is used to store cart detail**

Fieldname	Data type	Size	Constraint	Description
Cart_id	Int	10	Primary Key	To Store Cart_Id .
Category_Id	Int	25	Foreign Key	To Store Category_Id.
Item name	varchar	30	Not null	To Store Name of food item.
Item price	Decimal	8	Not null	To Store Price of food item.
Item quantity	Int	20	Not null	To Store Quantity of food item in the cart.
Total price	Decimal	8	Not null	To Store Total price for Quantity of food item in the cart.

Table Name : Order**Description : It is used to store Order details**

Fieldname	Data type	Size	Constraint	Description
Order_id	Int	10	Primary Key	To Store Order Id .
Cart_id	Int	20	Foreign Key	To Store Cart Id
Customer_id	Int	20	Not null	To Store Customer Id.
Orders_detail	varchar	50	Not null	To Store Order Details.
Order_total	Int	50	Not null	To Store Order Totals
Order_status	varchar	10	Not null	To Store Order Status

Table Name : Customer

Description : It is used to store Customer detail

Fieldname	Data type	Size	Constraint	Description
Customer_id	Int	10	Primary Key	To Store Customer Id .
Area_id	varchar	50	Foreign Key	To Store Customer Area
Customer_name	varchar	30	Not null	To Store Customer Name.
Email_id	varchar	30	Not null	To Store Email Id.
Password	varchar	15	Not null	To Store Password
Customer_address	varchar	100	Not null	To Store Customer Address
Customer_number	Int	10	Not null	To Store Customer Phone Number

Table Name : Customer_area

Description : It is used to store Customer area details

Fieldname	Data type	Size	Constraint	Description
Area_Id	Int	20	Primary Key	To Store Area Id
Name	Varchar	30	Not null	To Store Name of Area
City	Varchar	20	Not null	To Store Name of Area

Table Name : Feedback_form

Description : It is used to store Customer feedback

Fieldname	Data type	Size	Constraint	Description
Feedback_Id	Int	10	Primary Key	To Store the Feedback Id
Customer_Id	Int	10	Foreign Key	To Store Customer Id
Name	Varchar	30	Not null	To Store the Name of Customer
Feedback-date	DateTime	64	Not null	To Store Date and time when Feedback was sent.

Table Name : Payment**Description : It is used to store Payment details**

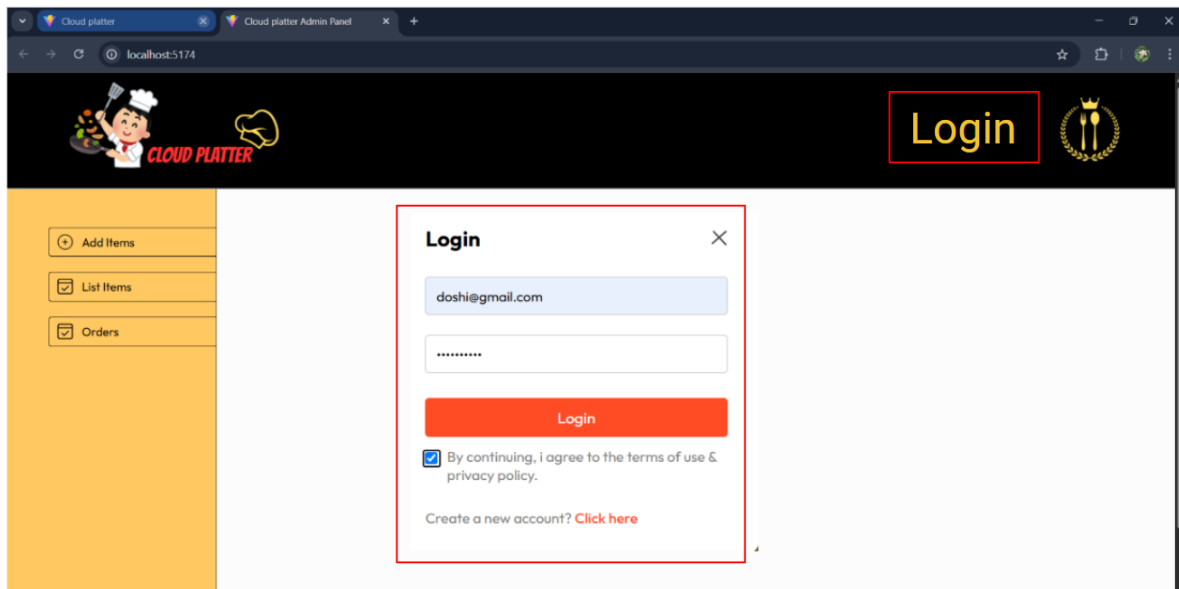
Fieldname	Data type	Size	Constraint	Description
Payment_Id	Int	20	Primary Key	To Store Payment Id .
Customer_Id	Int	10	Foreign Key	To Store Customer Id
Amount	Decimal	8	Not null	To Store the Amount.
Date	DateTime	64	Not null	To Store Date and time when Payment was sent.
Payment Method	Varchar	30	Not null	To Store type of Payment Method.
Status	Varchar	20	Not null	To Store the Status of Payment

Table Name : Contact_us**Description : It is used to store Contact details**

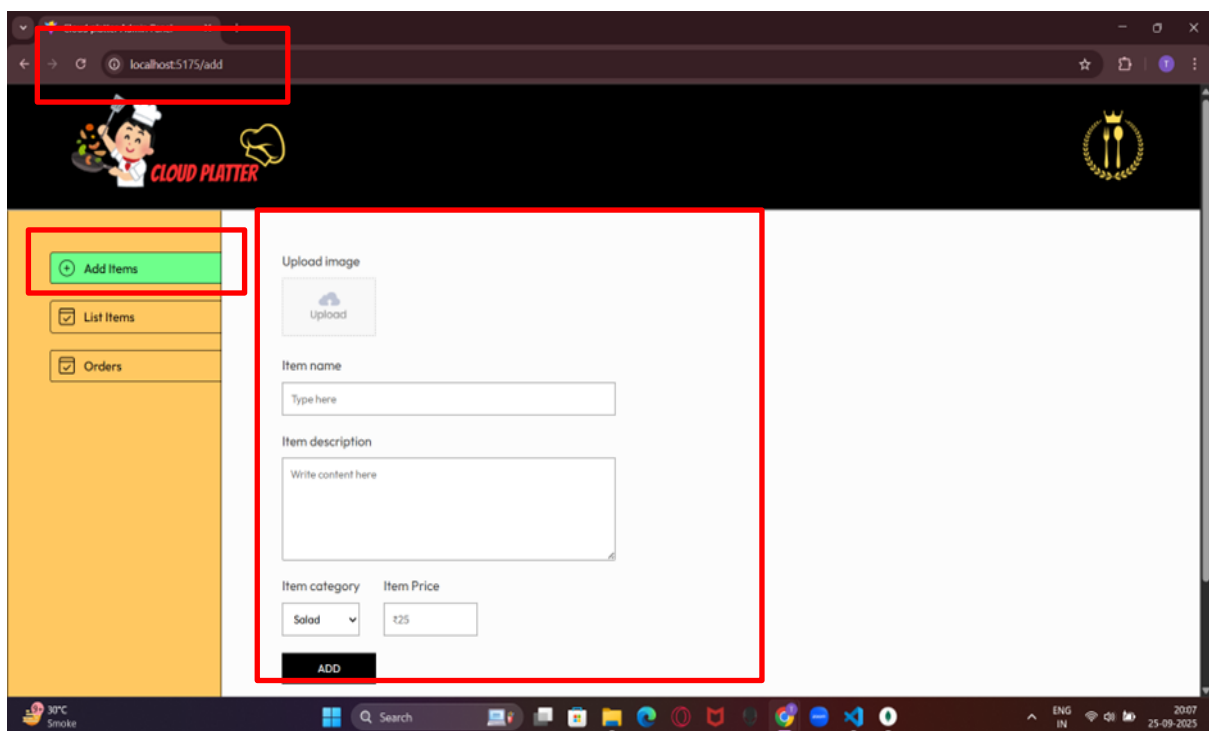
Fieldname	Data type	Size	Constraint	Description
Phone_Id	Int	10	Primary Key	To Store Phone_Id.
UserName	varchar	30	Not Null	To Store UserName .

Phone_number	Int	10	Not null	To Store Phone number.
Message	varchar	50	Not null	To Store Content of message.
Data_sent	DateTime	64	Not null	To Store Date and time when message was sent.

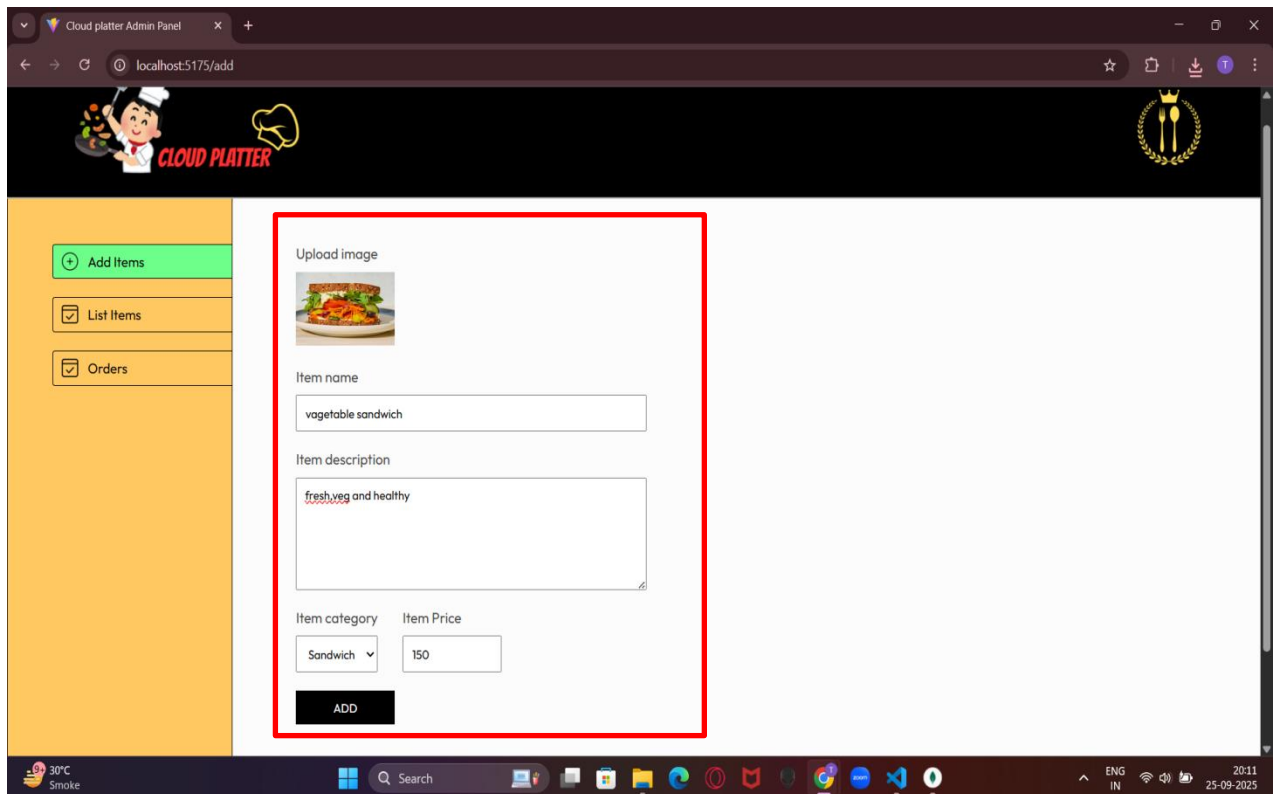
User Interface Screens



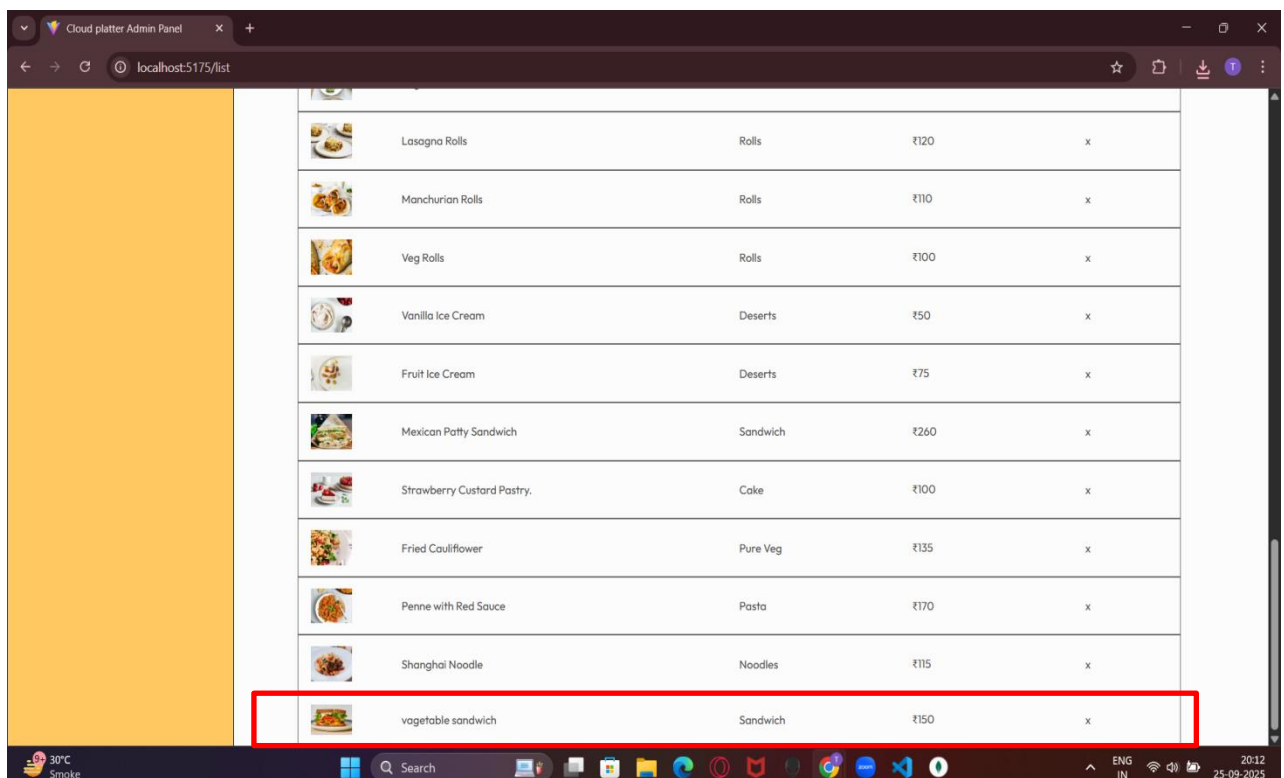
Description: It displays the Admin-side Login page.



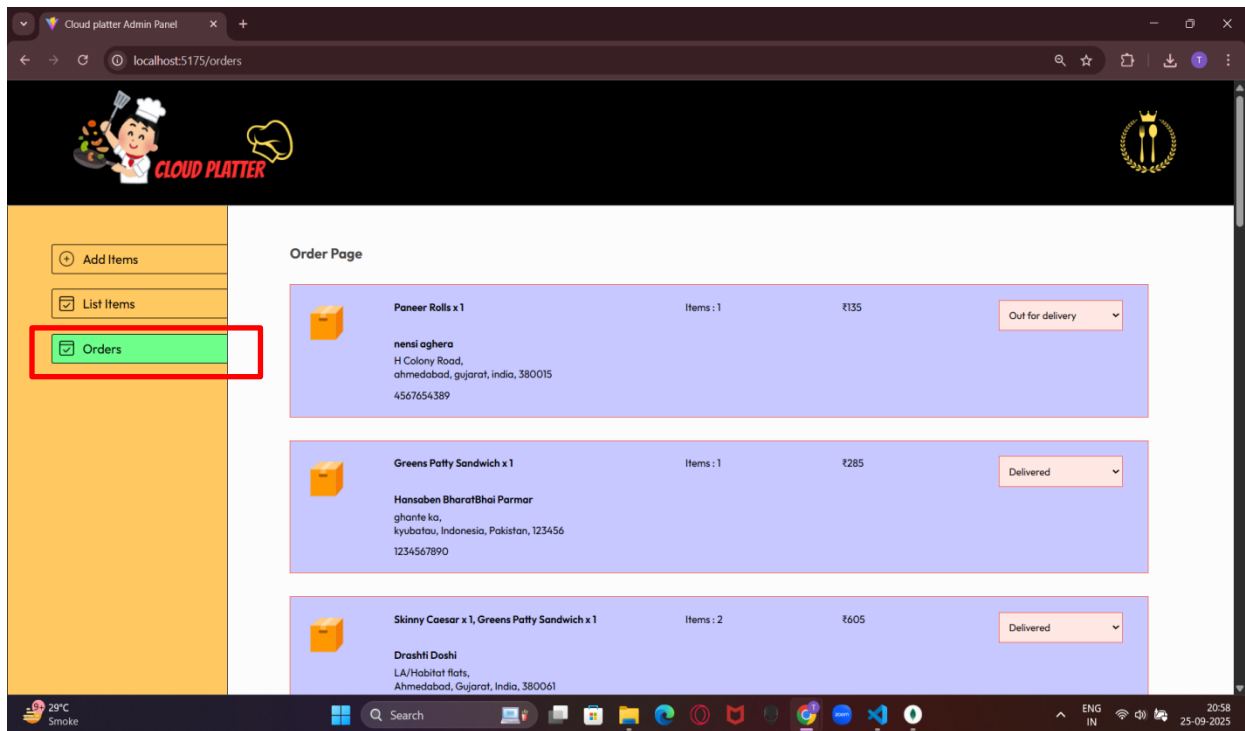
Description: It displays how Admin add's the new product.



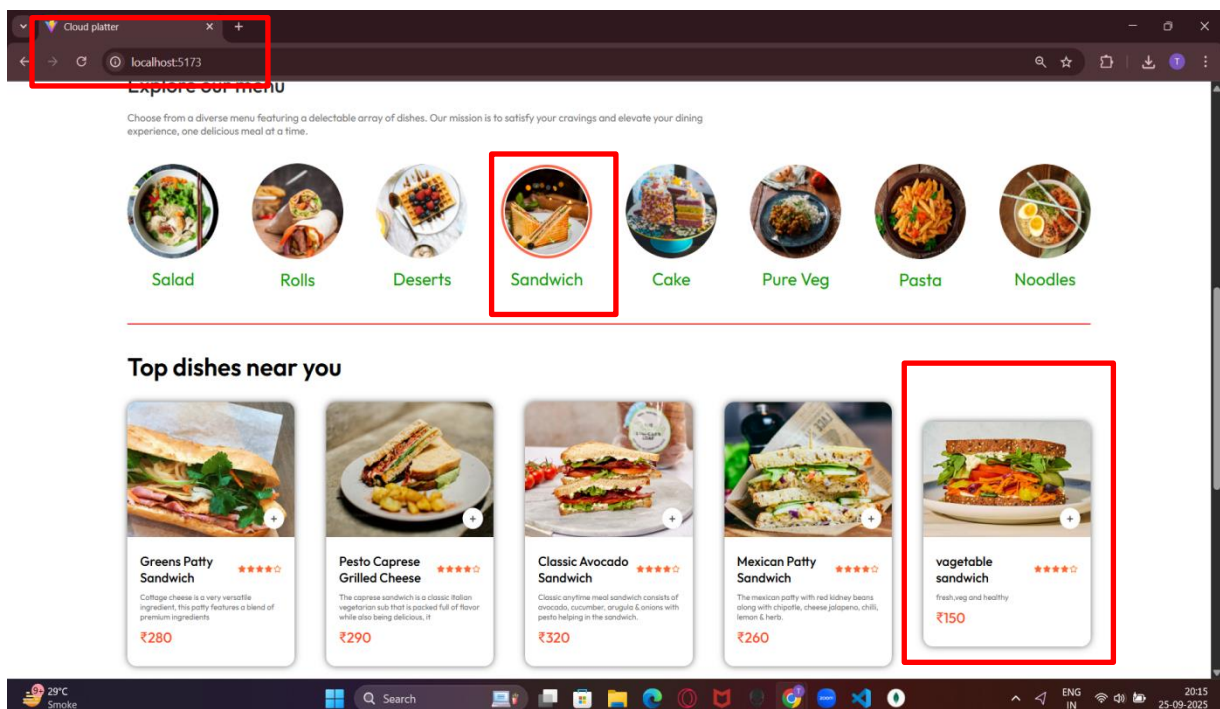
Description: It displays how Admin add's the new product.



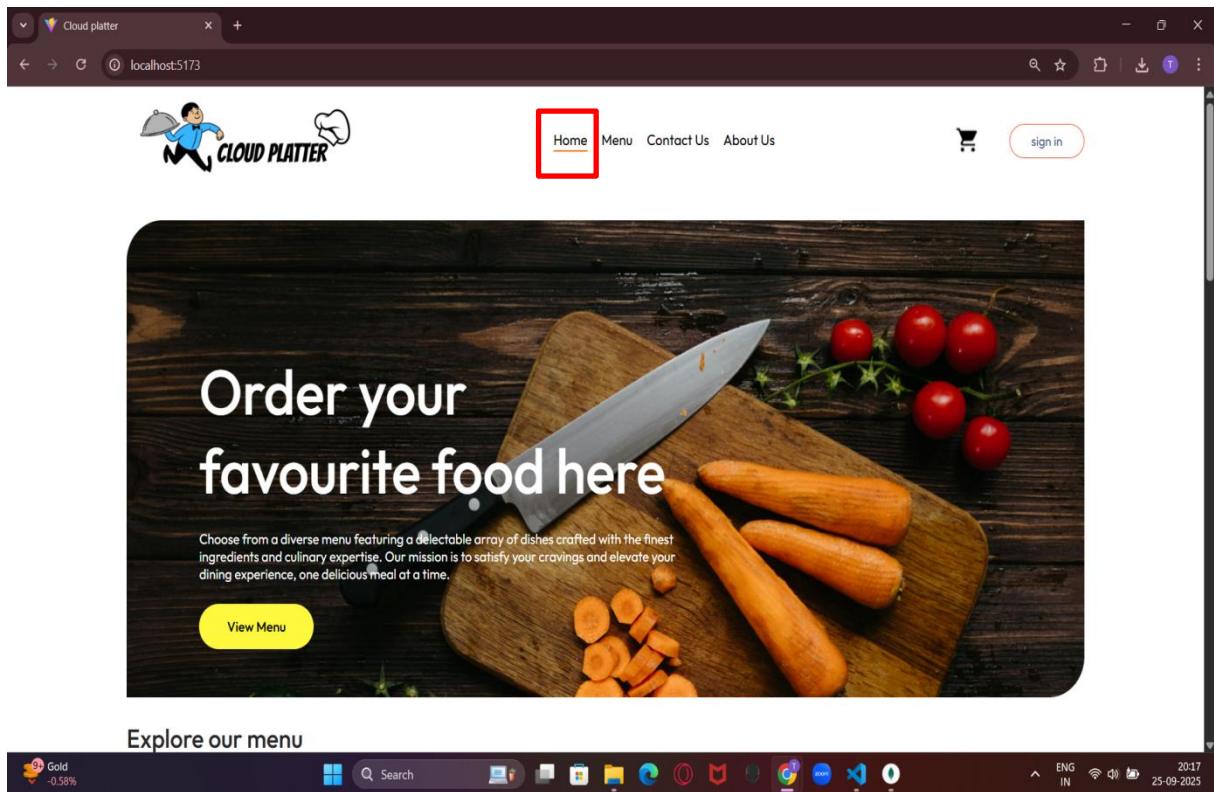
Description: It shows how Admin manages the order of the customer.



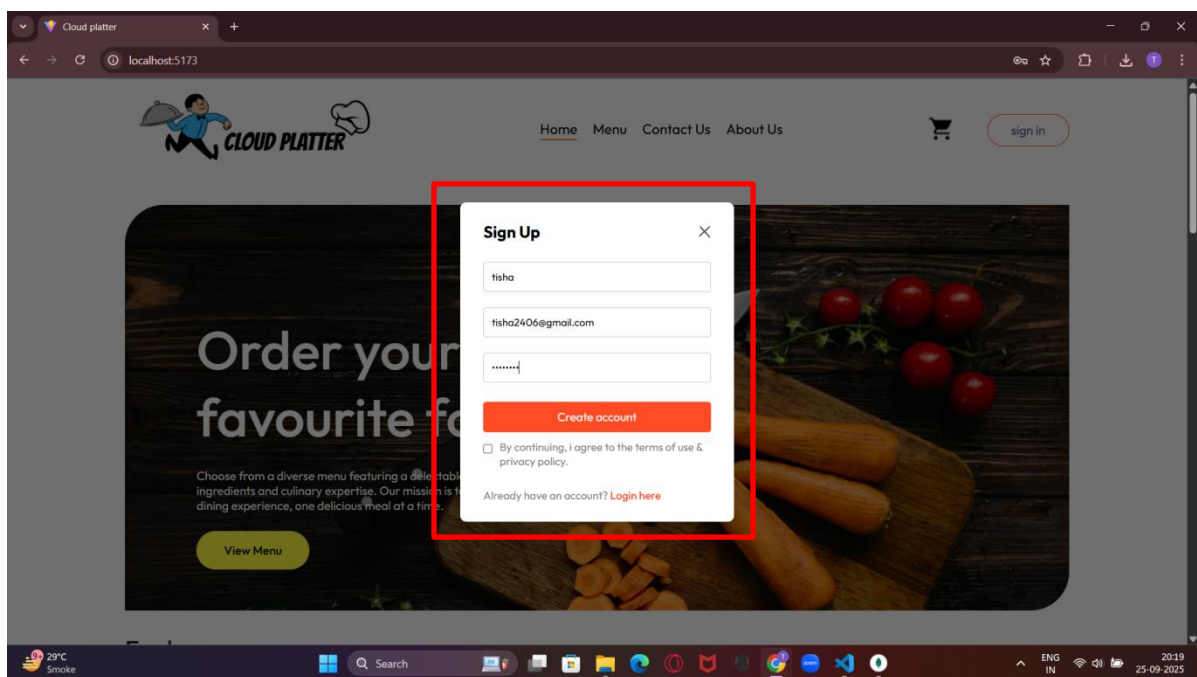
Description: It displays how Admin add's the new product.



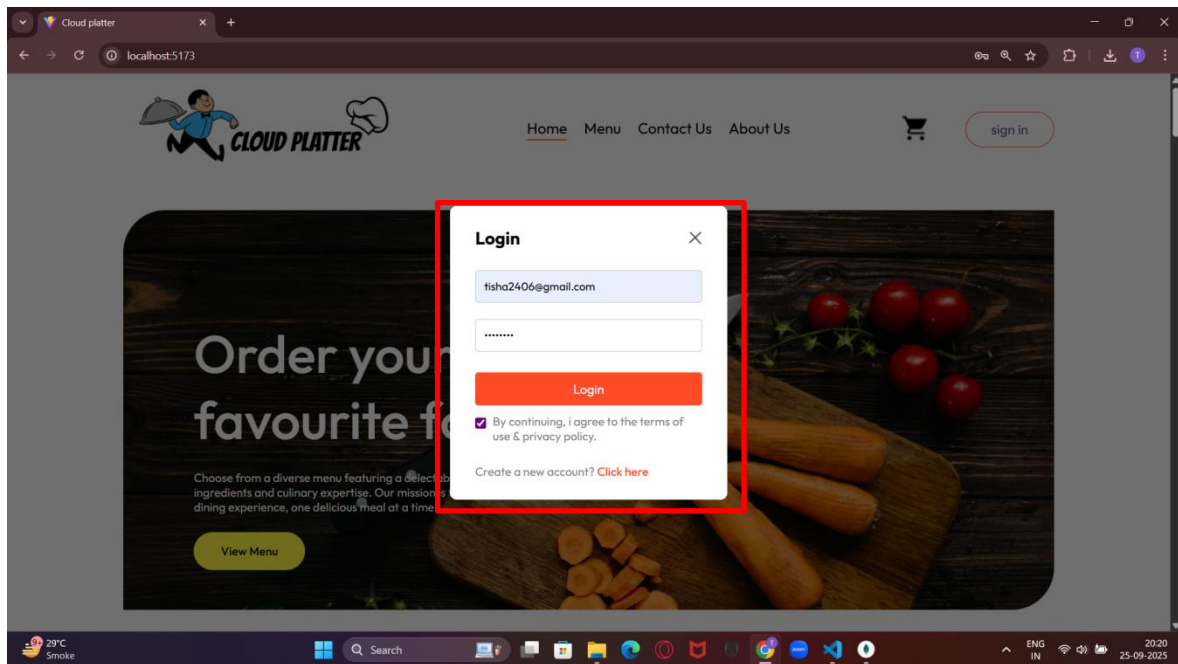
Description: This is Home page of our website



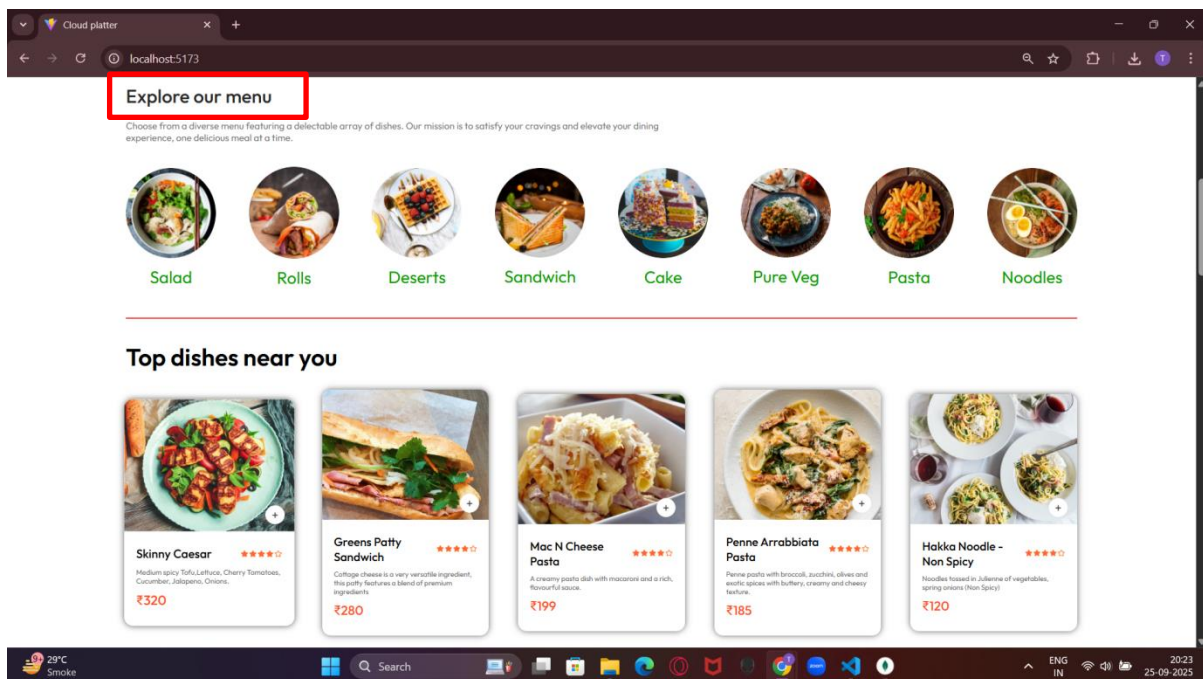
Description: If customer does not have an account they have to first Sign-up.



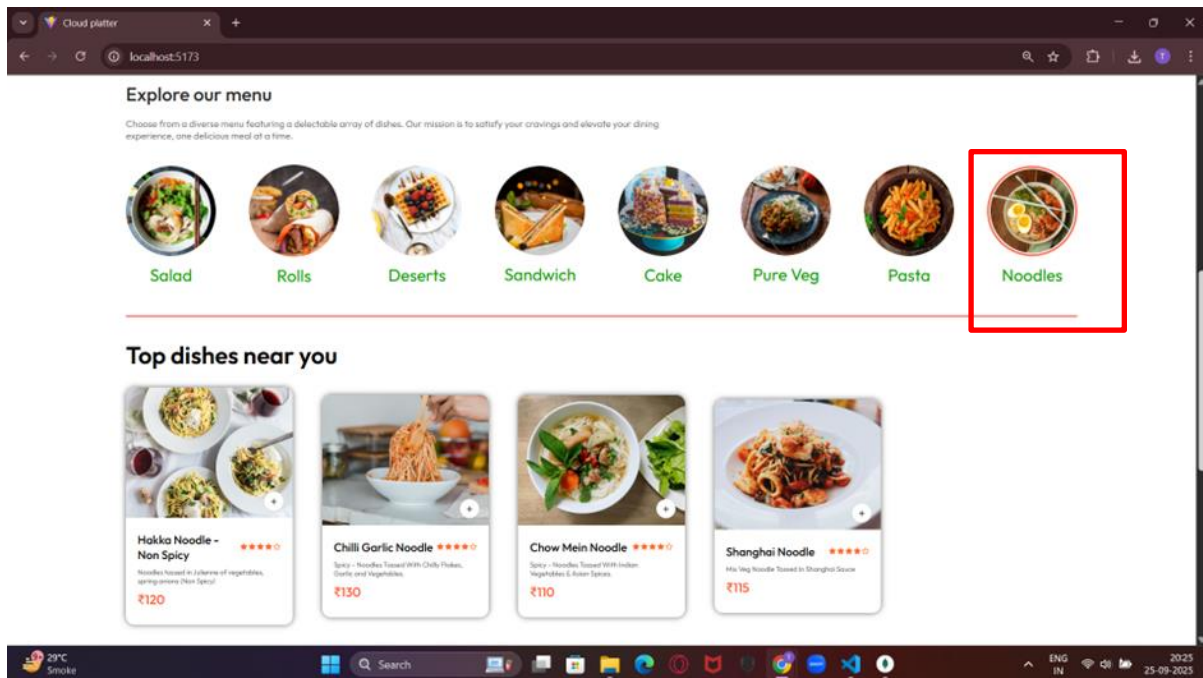
Description: If customer have an account then they have to just only Log-in.



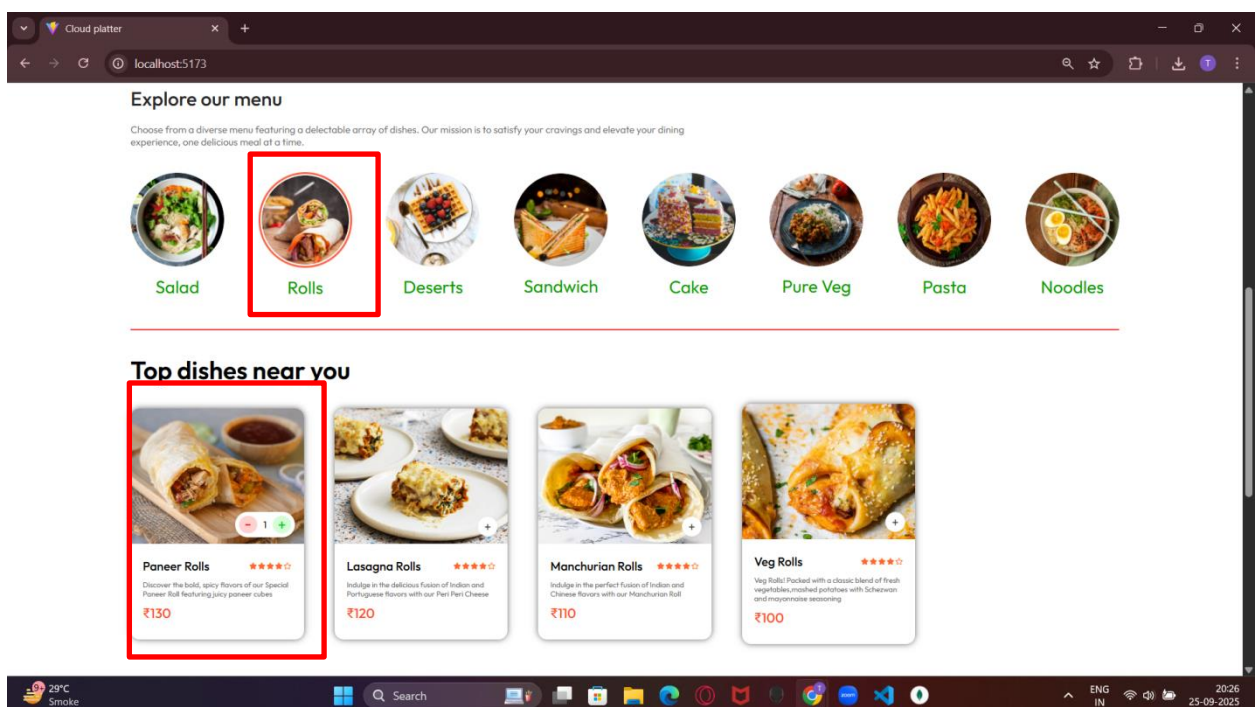
Description: It Shows the Categories of food items.



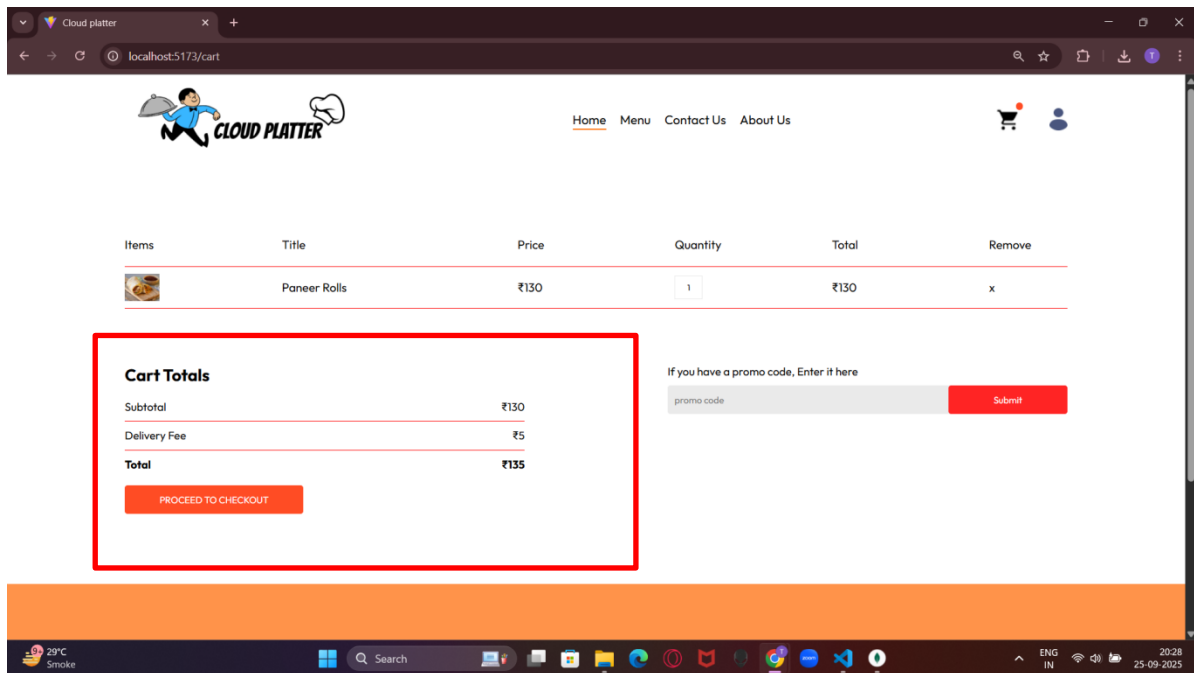
Description: It Shows the Sub-Category of any specific food item while clicking on it.



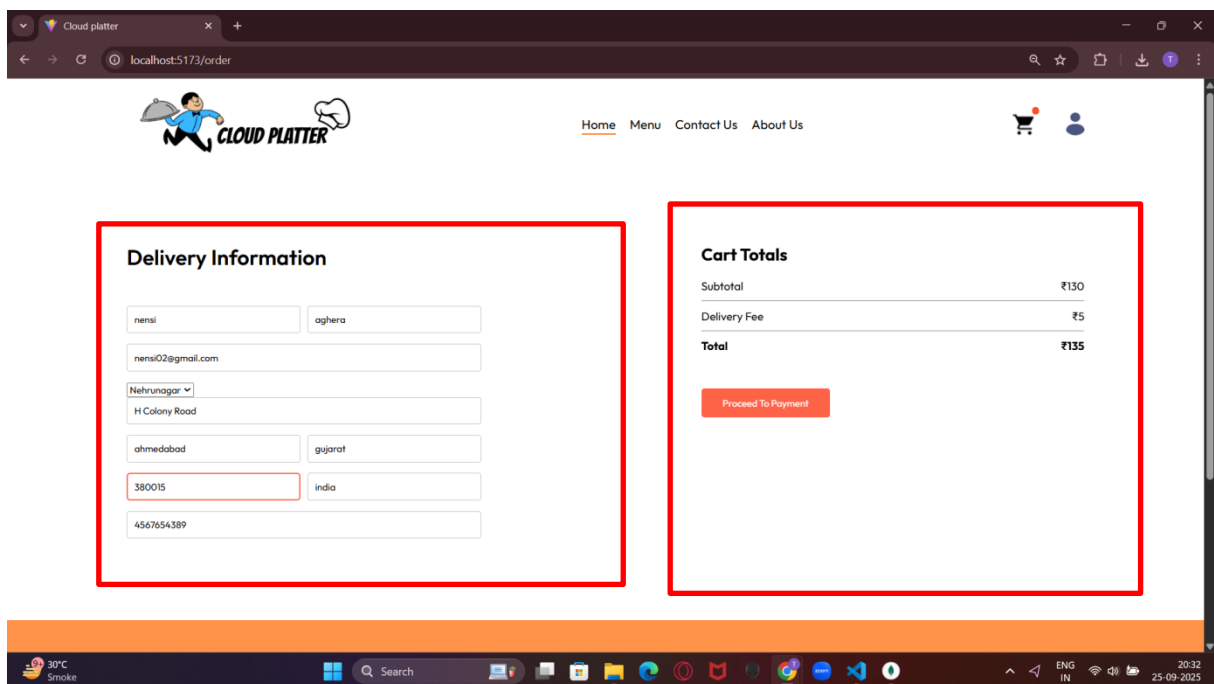
Description: It Shows the food item that is added to the cart.



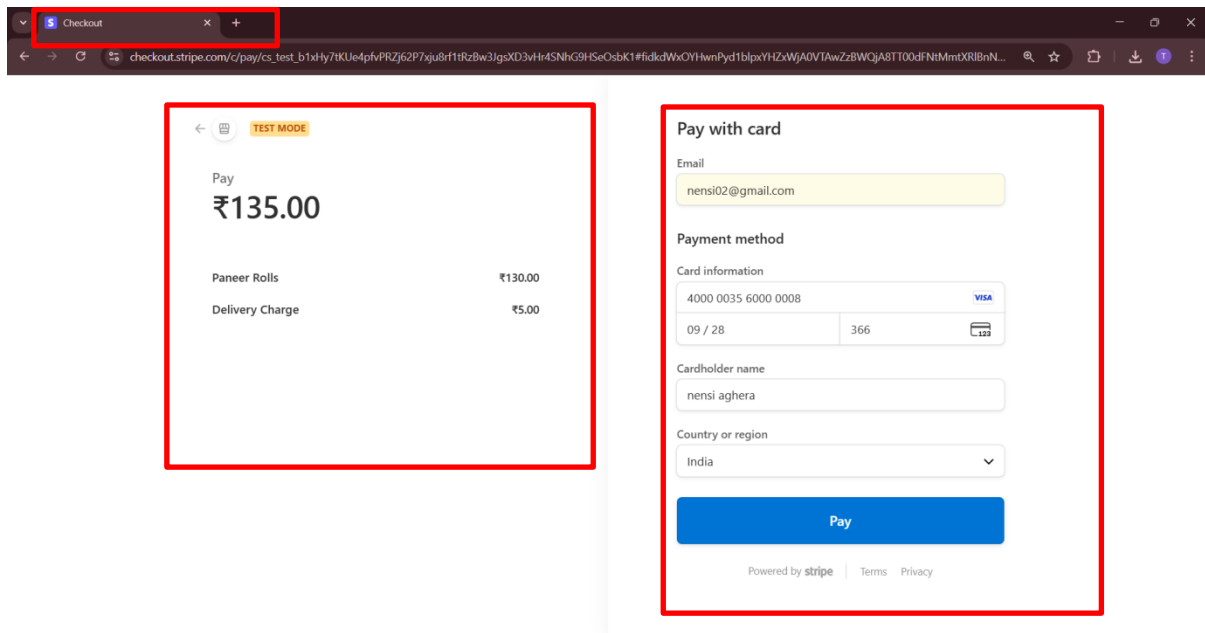
Description: It Shows the food item that is added to the cart.



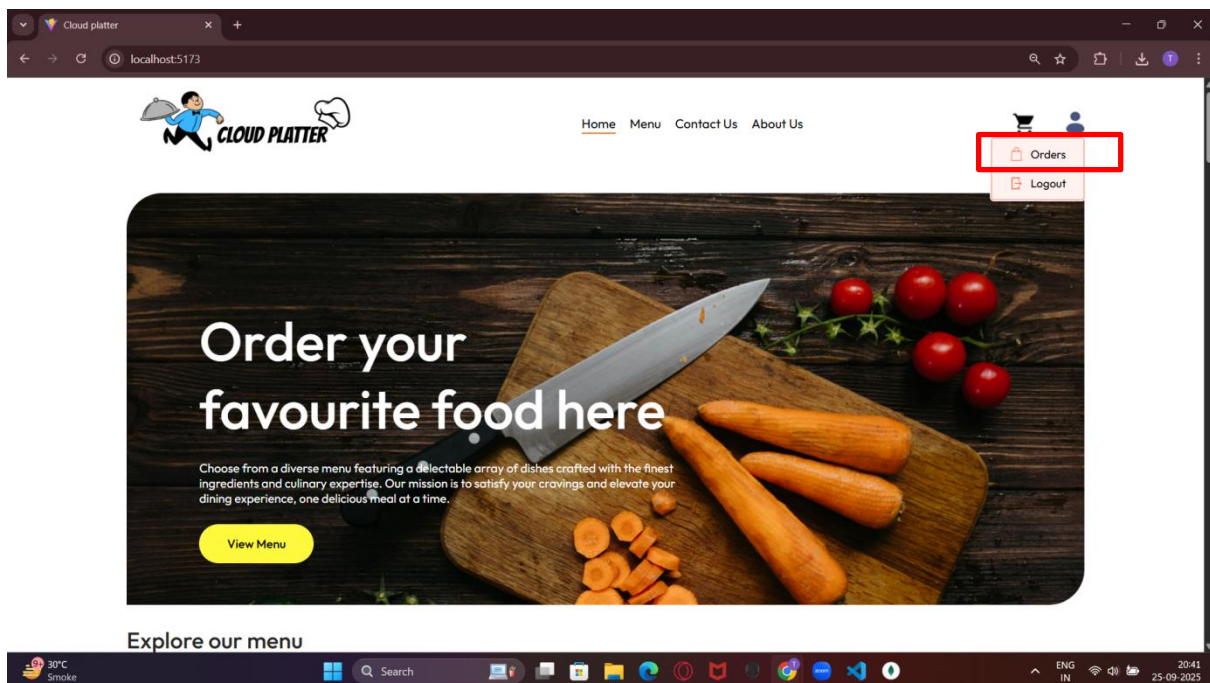
Description: It Shows the cart totals of food item which were added to the cart by customer.



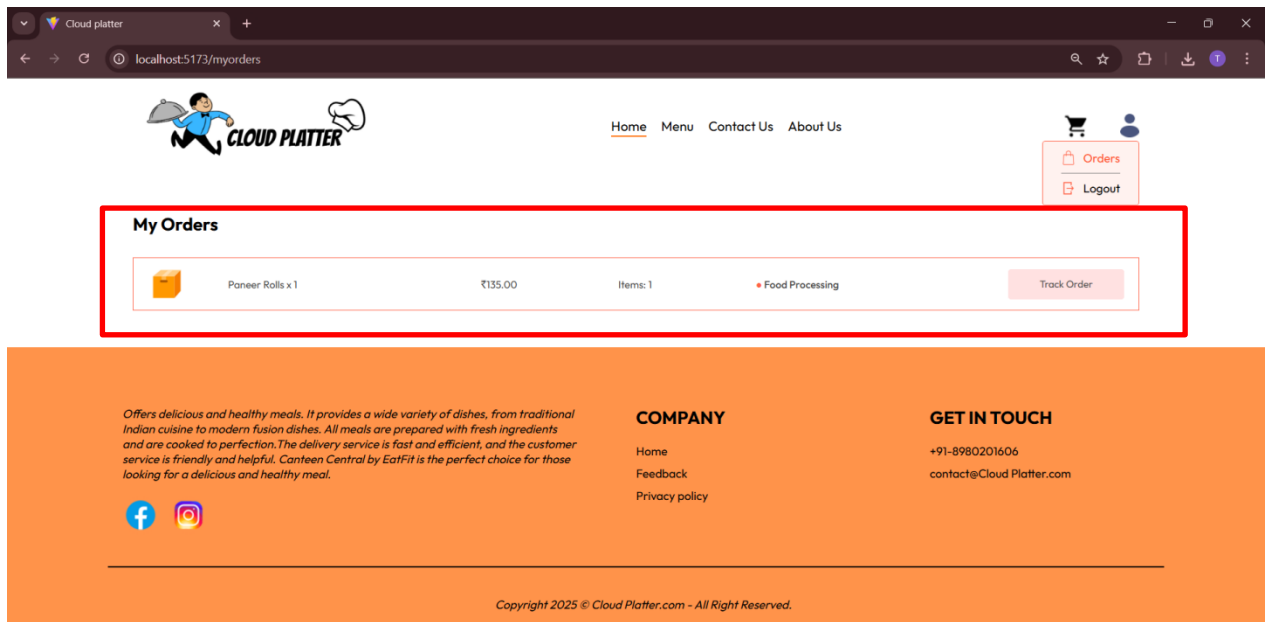
Description: It displays the Delivery information filled by customer for going further for proceed to payment side.



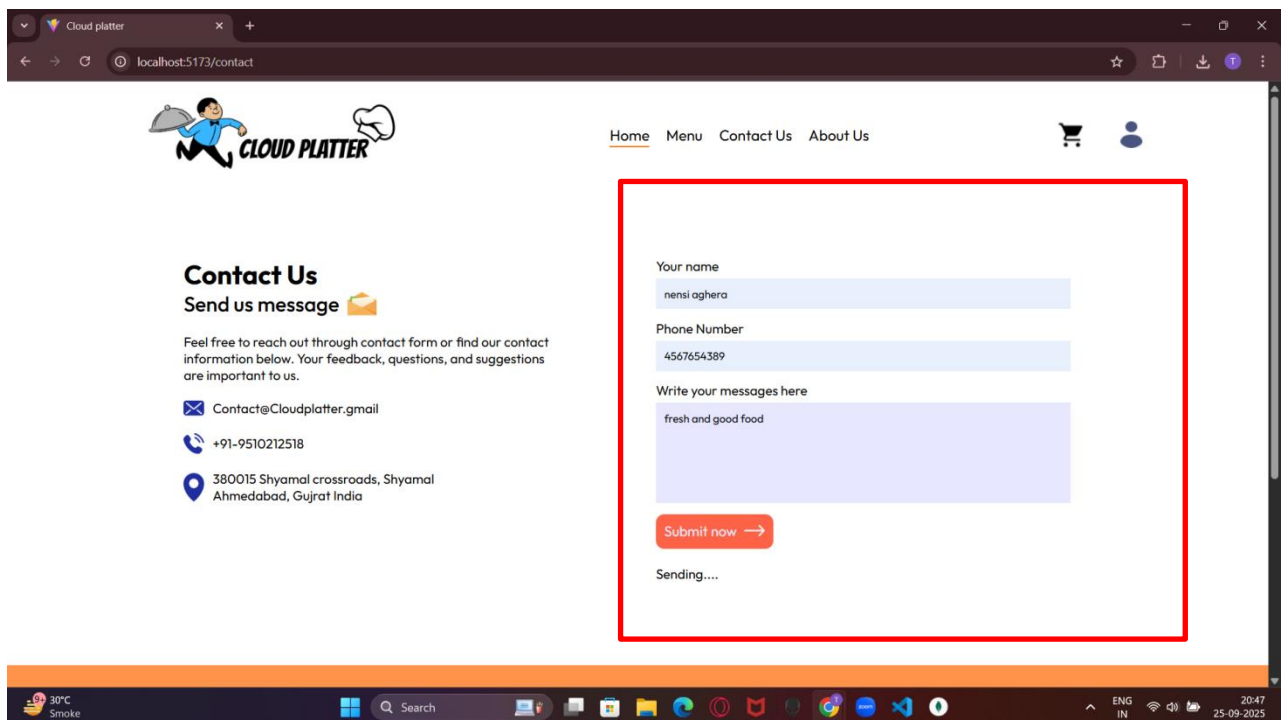
Description: It displays the current orders to the customer which is placed by him/her after completion of the payment.



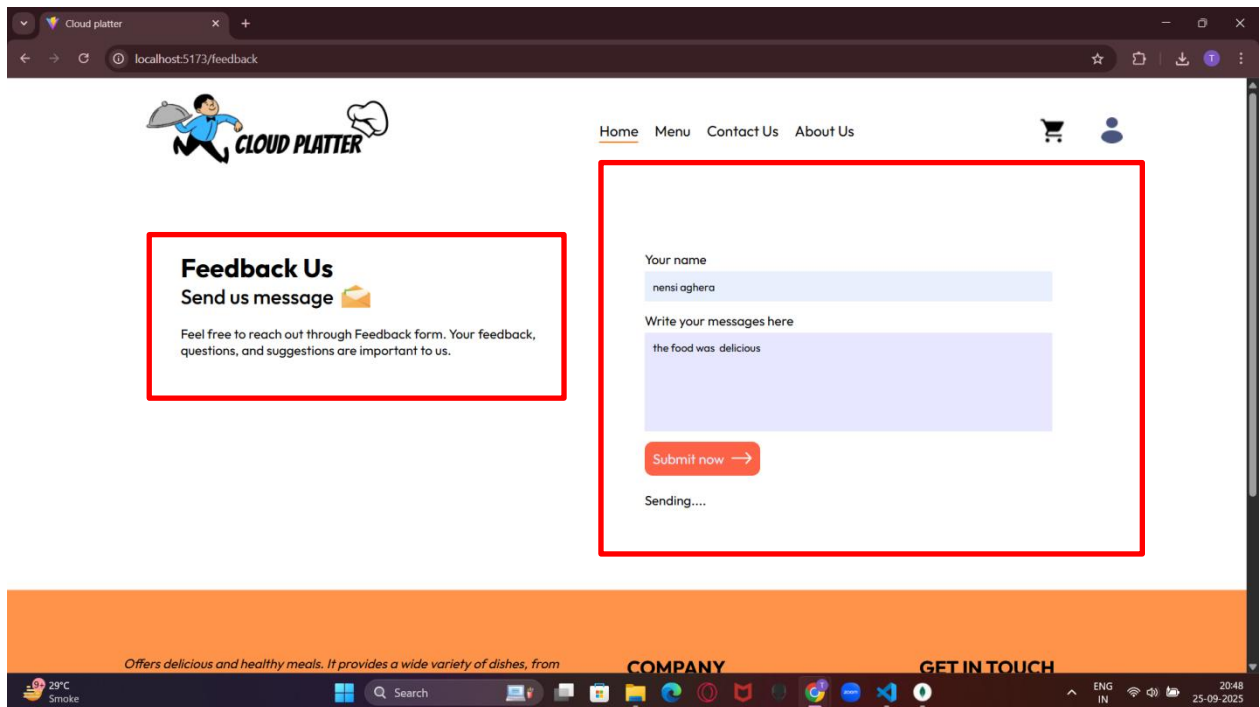
Description: It displays the current orders to the customer which is placed by him/her after completion of the payment.



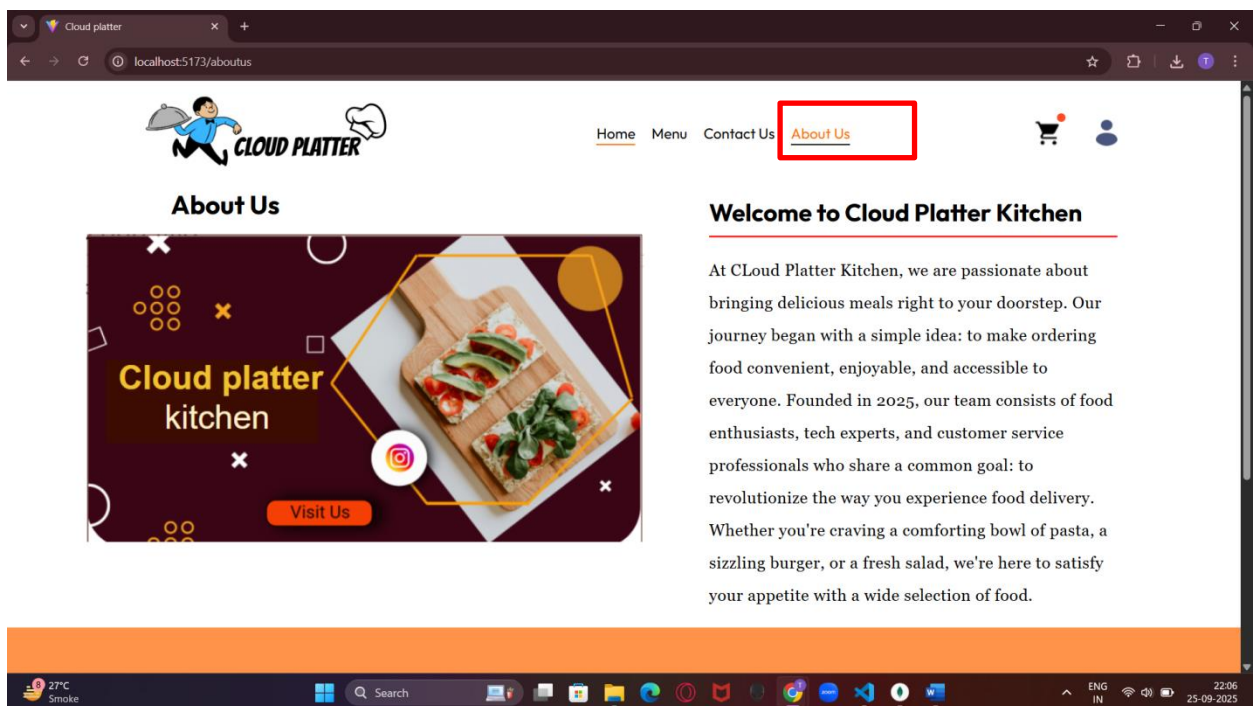
Description: It displays how customer can contact us through this.



Description: It displays how customer can give us feedback through this.



Description: This is a About-us page.



Reports

New Business

Home

Payments

Balances

Customers

Billing

Connect

More

Search

Payments

All payments

Collected fees

Disputes

All transactions

All6

Succeeded6

Refunded0

Uncaptured0

Failed0

Date & Time

Amount

Currency

Status

Payment method

More filters

Export

Edit columns

	Amount		Payment method	Description	Customer	Date		
☐	₹760.00	INR	Succeeded ✓	**** 0008	pi_3PB980SG5G59H81qVh0Ynmcqc	ompatel@gmail.com	Apr 30, 5:21 AM	...
☐	₹245.00	INR	Succeeded ✓	**** 0008	pi_3P91JVG59H81qVh0hOI0es5	heerrraakkk@gmail.com	Apr 26, 9:43 AM	...
☐	₹580.00	INR	Succeeded ✓	**** 0008	pi_3P5sF3SG59H81qVh0tU2rj0a	kahani@gmail.com	Apr 15, 4:18 PM	...
☐	₹364.00	INR	Succeeded ✓	**** 0008	pi_3P5qoG5G59H81qVh0SUm023P	parthiv@gmail.com	Apr 15, 2:47 PM	...
☐	₹22,720.00	INR	Succeeded ✓	**** 0008	pi_3P5kkmSG59H81qVh18tyPQRT	hitu912004@gmail.com	Apr 15, 8:19 AM	...
☐	₹85.00	INR	Succeeded ✓	**** 0008	pi_3P5RH2SG59H81qVh0B3ai0nX	hitu912004@gmail.com	Apr 14, 11:31 AM	...

6 results

Test mode

You're using test data. Activate your account to access all live data.

Activate account

Developers

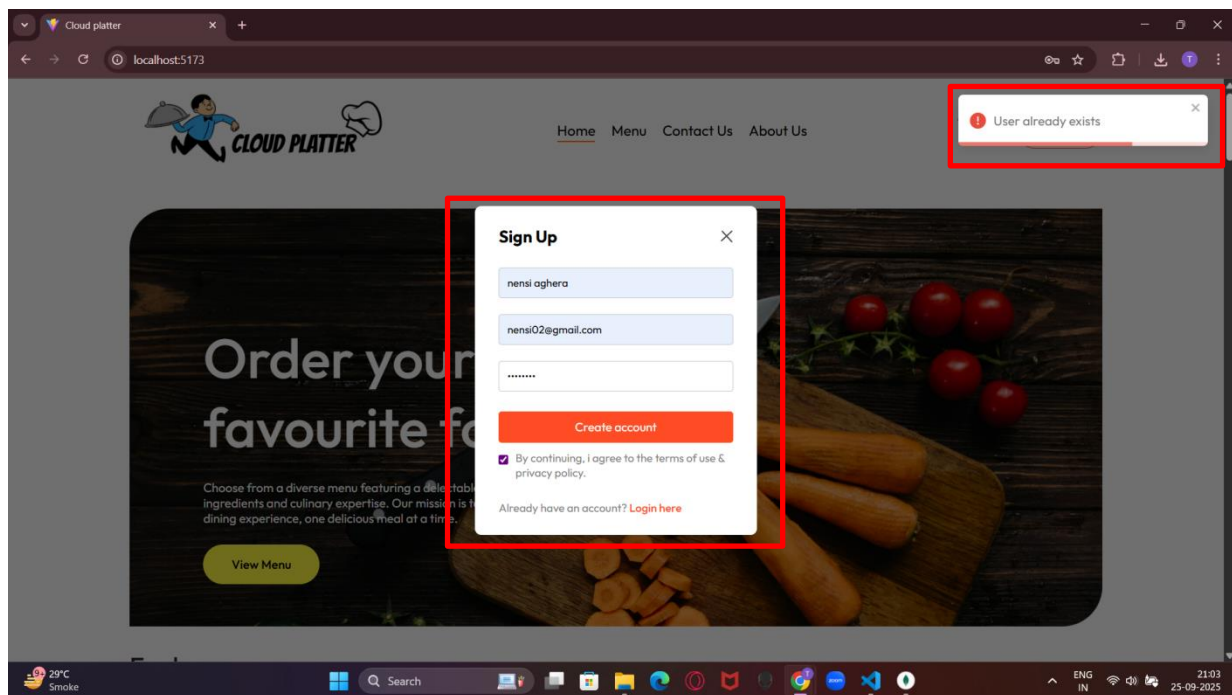
Test mode

Paused

Create payment

Test Plan and Test Cases

Test case Id	A001
Purpose	Customer registration
Test Created By	Nensi
Test Environment	Internet Explorer
Prerequisites	Enter User name, First name, last name, Email Id, Password etc.
Test Procedure	Check First name, last name etc.
Test Data	Username
Expected Result	If username is already existing then give error message.
Actual Result	User Registration Successful
Verdict: Pass/Fail	Pass
Comment	Registration For New Customer



Test case Id	A002
Purpose	Cart Page
Test Created By	Nensi
Test Environment	Internet Explorer
Prerequisites	Enter - Address, Email-id, Pin-code, Contact number etc.
Test Procedure	Check Email-id, Pin-code, Address etc.
Test Data	Email-id
Expected Result	If Email-id is not in correct format then give an error message.
Actual Result	Cart Details added Successful
Verdict: Pass/Fail	Pass
Comment	Address Details to Deliver the Food.

The screenshot shows a web browser window with the URL `localhost:5173/order`. The website is "Cloud Platter" and displays a "Delivery Information" form. The form fields are as follows:

- Name:** nensi (first name), anshu (last name)
- Email:** nensi02gmail.com (highlighted with a red box and an error message: "Please include an '@' in the email address. 'nensi02gmail.com' is missing an '@'").
- Address:** H Colony Road
- City:** ahmedabad
- State:** gujarat
- Pin Code:** 380015
- Country:** india
- Phone Number:** 4567654389

On the right side, the "Cart Totals" are displayed:

Cart Totals	
Subtotal	₹130
Delivery Fee	₹5
Total	₹135

Below the cart totals is a red button labeled "Proceed To Payment".

Test case Id	A003
Purpose	Feedback details
Test Created By	Nensi
Test Environment	Internet Explorer
Prerequisites	Enter - First name, Last name, Phone number etc.
Test Procedure	First name, last name, etc
Test Data	Phone number
Expected Result	If Phone number is empty then give an error message
Actual Result	Contact Details Successful
Verdict: Pass/Fail	Pass
Comment	If a customer has any query.

Cloud platter

localhost:5173/contact

Home Menu Contact Us About Us

Contact Us

Send us message

Feel free to reach out through contact form or find our contact information below. Your feedback, questions, and suggestions are important to us.

Contact@Cloudplatter.gmail

+91-9510212518

380015 Shyamal crossroads, Shyamal
Ahmedabad, Gujrat India

Your name
nensi aghera

Phone Number
Enter your mobile number

Write your messages
tasty and good food

Please fill out this field.

Submit now →

Show desktop

28°C Smoke

Search

ENG IN

21:08
25-09-2023

Future Enhancement

In our continuous commitment to innovation and excellence, we envision enhancing our platform with state-of-the-art features such as AI-powered recommendation systems for personalized menu suggestions, integration of IoT devices for real-time kitchen monitoring and optimization, and implementation of blockchain technology for enhanced food traceability and transparency, ensuring that we stay at the forefront of culinary technology and continue to exceed our customers' expectations.

Some are the expansion which we have thought of are :

- Try to make system more user friendly.
- Send verification code for login and registration on mobile by SMS.
- AI-Powered Menu Recommendations

References

- React Official Docs <https://react.dev/reference/react>
- W3Schools <https://www.w3schools.com/>
- Javatpoint <https://www.javatpoint.com/>
- Stack Overflow <https://stackoverflow.com>
- Learning React <https://www.amazon.com/dp/1492051721>
- Codevolution <https://www.youtube.com/c/Codevolution>
- The Net Ninja <https://www.youtube.com/c/TheNetNinja>
- Draw.io <https://app.diagrams.net/>